

추천 시스템

추천 시스템 1

1. 추천 시스템 이름→ "카테고리별 인기 상품"

2. 추천 시스템의 테마: 추천 시스템의 고유 컨셉에 대한 설명

고객이 카테고리의 인기 상품을 바로 확인할 수 있도록,
категори를 나누고 평점 수와 평점을 결합한 추천을 제공합니다.

3. 구현 로직: SQL 쿼리 설명 및 주요 로직 설명

rating_count이 많은 상품을 판매가 많은 상품으로 간주했습니다.

rating_count이 같다면, **rating**이 높은 순으로 정렬했습니다.

카테고리가 약 4덱스 정도로, 3덱스로 설정시 163개, 2덱스 설정 시 70개의 상품이 나왔습니다.

총 상품 갯수가 1459개로, 3덱스로 설정한다면 약 상위 11% 상품이 추천되기에 설정했습니다.

```
WITH category_base AS (
  SELECT
    product_id,
    product_name,
    rating_count,
    SPLIT(category, '|')[SAFE_OFFSET(2)] AS category_level3,
    SAFE_CAST(rating AS FLOAT64) AS rating_num,
  FROM `amazonrecomm.new.recomm`
),
ranked AS (
  SELECT
    category_level3,
    product_id,
    product_name,
    rating_num,
    rating_count,
    ROW_NUMBER() OVER (
      PARTITION BY category_level3
      ORDER BY rating_count DESC, rating_num DESC
    ) AS rn
  FROM category_base
  WHERE category_level3 IS NOT NULL
    AND rating_num IS NOT NULL
)
SELECT
  category_level3,
  product_id,
  product_name,
  rating_num,
  rating_count
FROM ranked
WHERE rn <= 3
ORDER BY category_level3, rn
```

4. 결과

카테고리 3덱스별 평점이 높고, 평점수가 높은 상위 3개 상품을 보여줍니다.

Query results						
+ Create conversation Save results Open in						
Job information						
Results						
Visualization						
JSON						
Execution details						
Execution graph						
Row	category_level3	product_id	product_name	rating_num	rating_count	
1	AVReceivers&Amplifiers	B0978V2CP6	Cubetek 3 in 1 LCD Display V5.0 Bluetooth Transmitter Receiver, Bypass Audio Adapter with Aux, Optical, Dual Link Support for TV, Home Stereo, PC, Headphones, Speakers, Model: CB-BT27	4.0	897	
2	Accessories	B014I8SX4Y	Amazon Basics High-Speed HDMI Cable, 6 Feet (2-Pack),Black	4.4	426973	
3	Accessories	B014I8SSD0	Amazon Basics High-Speed HDMI Cable, 6 Feet - Supports Ethernet, 3D, 4K video,Black	4.4	426973	
4	Accessories	B07KSMBL2H	AmazonBasics Flexible Premium HDMI Cable (Black, 4K@60Hz, 18Gbps), 3-Foot	4.4	426973	
5	Adapters	B08BCKN299	Source Gold Plated 3.5 mm Headphone Splitter for Computer 2 Male to 1 Female 3.5mm Headphone	3.9	6491	
6	Adapters	B08BCKN299	Mic Audio Y Splitter Cable Smartphone Headset to PC Adapter - (Black,20cm)	3.9	6491	
7	Adapters	B097C564GC	Source Gold Plated 3.5 mm Headphone Splitter for Computer 2 Male to 1 Female 3.5mm Headphone	3.9	6491	
8	Adapters	B07WKB03...	Mic Audio Y Splitter Cable Smartphone Headset to PC Adapter - (Black,20cm)	4.3	4426	
9	AirConditioners	B09NS5TKPN	rt's [2 Pack] Mini USB C Type C Adapter Plug, Type C Female to USB A Male Charger Charging Cable	4.0	2581	
10	AirPurifiers	B01L6MT7E0	Adapter Converter compatible for iPhone, Samsung S20 ultra/S21/S10/S8/S9/MacBook Pro iPad Silver	4.3	3231	
11	AirPurifiers	B0811V0GJ5	ESnipe Mart Worldwide Travel Adapter with Build in Dual USB Charger Ports with 125V 6A, 250V Protected Electrical Plug for Laptops, Cameras (White)	4.1	4157	

추천 시스템 2

1. 추천 시스템 이름→ "가성비 추천"

2. 추천 시스템의 테마: 추천 시스템의 고유 컨셉에 대한 설명

저렴하면서 평점이 높은 상품을 추천하여 고객의 선택을 돕는다.

3. 구현 로직: SQL 쿼리 설명 및 주요 로직 설명

동일 카테고리에서 가성비 점수를 계산하였습니다.

가성비 점수는 평점이 높으면 점수가 커지게, 가격은 낮으면 점수가 커지게 **평점/가격**으로 설정했습니다.

다만 초저가 상품이 높은 점수를 받는 것을 방지하고자 SQRT 를 사용하여 가격 차이 영향을 조정했습니다.

그럼에도 가격이 저렴한 10불 이하의 상품은 매출에 영향이 적을 것으로 보고 제외했습니다.

또한 신뢰도를 위해 평점의 수가 50개 미만인 상품도 제외했습니다.

```
WITH base AS (
  SELECT
    product_id,
    product_name,
    SPLIT(category, '|')[SAFE_OFFSET(2)] AS category_level3,
    discounted_price,
    SAFE_CAST(rating AS FLOAT64) AS rating_num,
    rating_count
  FROM `amazonrecomm.new.recomm`
),
ranked AS (
  SELECT
    category_level3,
    product_id,
    product_name,
    discounted_price,
    rating_num,
    rating_count,
    value_score,
    ROW_NUMBER() OVER (
      PARTITION BY category_level3
      ORDER BY value_score DESC, rating_num DESC, rating_count DESC
    ) AS rn
  FROM scored
```

```

)
SELECT
    category_level3,
    product_id,
    product_name,
    discounted_price,
    rating_num,
    rating_count,
    value_score
FROM ranked
WHERE rn <= 3
ORDER BY category_level3, rn

```

4. 결과

카테고리 별 가성비 상품 3개를 보여줍니다.

Query results								
+ Create conversation Save results Open in								
Job information Results Visualization JSON Execution details Execution graph								
Row	category_level3	product_id	product_name	discounted_price	rating_num	rating_count	value_score	
1	AVReceivers&Amplifiers	B0978V2CP6	Cubetek 3 in 1 LCD Display V5.0 Bluetooth Transmitter Receiver, Bypass Audio Adapter with Aux, Optical, Dual Link Support for TV, Home Stereo, PC, Headphones, Speakers, Model: CB-BT27	1990.0	4.0	897	0.08967	
2	Accessories	B07VVXJ2P5	SVM Products Unbreakable Set Top Box Stand with Dual Remote Holder (Black)	96.0	3.6	1796	0.36742	
3	Accessories	B08PPHFXG3	Posh 1.5 Meter High Speed Gold Plated HDMI Male to Female Extension Cable (Black)	173.0	4.3	1237	0.32692	
4	Accessories	B06XR9PR5X	Amazon Basics HDMI Coupler,Black	209.0	4.4	18872	0.30435	
5	Adapters	B08BCKN299	Sounce Gold Plated 3.5 mm Headphone Splitter for Computer 2 Male to 1 Female 3.5mm Headphone Mic Audio Y Splitter Cable Smartphone Headset to PC Adapter - (Black,20cm)	120.0	3.9	6491	0.35602	
6	Adapters	B08BCKN299	Sounce Gold Plated 3.5 mm Headphone Splitter for Computer 2 Male to 1 Female 3.5mm Headphone Mic Audio Y Splitter Cable Smartphone Headset to PC Adapter - (Black,20cm)	120.0	3.9	6491	0.35602	
7	Adapters	B094DQWV9B	Kanget [2 Pack] Type C Female to USB A Male Charger Charging Cable Adapter Converter compatible for iPhone 14, 13, 12, 11 Pro Max/Mini/XR/XS/X/SE, Samsung S20 ultra/S21/S10/S8/S9/MacBook Pro iPad (Grey)	149.0	4.0	1540	0.32769	
8	Adapters&Multi-Outlets	B07WKBD3...	ESnipe Mart Worldwide Travel Adapter with Build in Dual USB Charger Ports with 125V 6A, 250V Protected Electrical Plug for Laptops, Cameras (White)	425.0	4.0	2581	0.19403	
9	AirConditioners	B09NSSTKPN	LG 1.5 Ton 5 Star AI DUAL Inverter Split AC (Copper, Super Convertible 6-in-1 Cooling, HD Filter with Anti-Virus Protection, 2022 Model, PS-Q19YNZE, White)	42990.0	4.3	3231	0.02074	

추천 시스템 3

1. 추천 시스템 이름→ "프리미엄 추천"

2. 추천 시스템의 테마: 추천 시스템의 고유 컨셉에 대한 설명

가격이 높은 상품을 우선 선정하고, 그중 평점과 리뷰 수가 높은 상품을 추천했습니다.

3. 구현 로직: SQL 쿼리 설명 및 주요 로직 설명

동일 카테고리에서 가격이 높은 상품, 평점이 높은 순, 평점이 많은 순으로 정렬했습니다.

신뢰도를 위해 평점의 수는 50개 미만은 제외했습니다.

프리미엄인 만큼, 평점도 4.0 이상으로 설정했습니다.

rank 의 base는 위에 만든 base 를 사용했습니다.

```

ranked AS (
    SELECT
        category_level3,
        product_id,
        product_name,
        discounted_price,
        rating_num,
        rating_count,
        ROW_NUMBER() OVER (

```

```

PARTITION BY category_level3
ORDER BY discounted_price DESC, rating_num DESC, rating_count DESC
) AS rn
FROM base
WHERE category_level3 IS NOT NULL
      AND rating_num IS NOT NULL
      AND rating_count > 50
      AND rating_num >= 4.0
)

SELECT
  category_level3,
  product_id,
  product_name,
  discounted_price,
  rating_num,
  rating_count,
  value_score
FROM ranked
WHERE rn <= 3
ORDER BY category_level3, rn

```

4. 결과

카테고리별 프리미엄 상품3개를 보여줍니다.

다만 평점 4.0이상으로 조건이 걸려서, 3번 가성비 추천(161개)보다 143개로 적습니다.

Query results							
Job information		Results	Visualization	JSON	Execution details	Execution graph	
Row	category_level3	product_id	product_name	discounted_price	rating_num	rating_count	
1	AVReceivers&Amplifiers	B0978V2CP6	Cubetek 3 in 1 LCD Display V5.0 Bluetooth Transmitter Receiver, Bypass Audio Adapter with Aux, Optical, Dual Link Support for TV, Home Stereo, PC, Headphones, Speakers, Model: CB-BT27	1990.0	4.0	897	
2	Accessories	B09HN7LD5L	PROLEGEND® PL-T002 Universal TV Stand Table Top for Most 22 to 65 inch LCD Flat Screen TV, VESA up to 800 by 400mm	1850.0	4.0	184	
3	Accessories	B07966M8XH	Model-P4 6 Way Swivel Tilt Wall Mount 32-55-inch Full Motion Cantilever for LEDLCD and Plasma TVs	1599.0	4.2	2727	
4	Accessories	B074CWD7MS	Digitex DTR 550 LW (67 inch) Tripod For DSLR, Camera Operating Height: 5.57 Feet Maximum Load Capacity up to 4.5kg Portable Lightweight Aluminum Tripod with 360 Degree Ball Head Carry Bag Included (Black) (DTR 550LW)	1549.0	4.4	15137	
5	Adapters	B097C564GC	rt's [2 Pack] Mini USB C Type C Adapter Plug, Type C Female to USB A Male Charger Charging Cable Adapter Converter compatible for iPhone, Samsung S20 ultra/S21/S10/S8/S9/MacBook Pro iPad Silver	294.0	4.3	4426	
6	Adapters	B094DQWV9B	Kanget [2 Pack] Type C Female to USB A Male Charger Charging Cable Adapter Converter compatible for iPhone 14, 13, 12, 11 Pro Max/Mini/XR/XS/X/SE, Samsung S20 ultra/S21/S10/S8/S9/MacBook Pro iPad (Grey)	149.0	4.0	1540	
7	Adapters&Multi-Outlets	B07WKBD37W	ESnipe Mart Worldwide Travel Adapter with Build in Dual USB Charger Ports with 125V 6A, 250V Protected Electrical Plug for Laptops, Cameras (White)	425.0	4.0	2581	
8	AirConditioners	B09NSSTKPN	LG 1.5 Ton 5 Star AI DUAL Inverter Split AC (Copper, Super Convertible 6-in-1 Cooling, HD Filter with Anti-Virus Protection, 2022 Model, PS-Q19YNZE, White)	42990.0	4.3	3231	
9	AirPurifiers	B01KCSGBU2	Philips Air Purifier Ac2887/20,VitaShield Intelligent Purification,Long Hepa Filter Life Up to 17000 Hours,Removes 99.9% Airborne Viruses & Bacteria,99.97% Airborne Pollutants,Ideal For Master	14499.0	4.3	2026	